dp练习

人员

杨咏丞、刘宸熙、李瑞涵、田心一、柳力玮、初锦阳、刘祺、纪博涵、苑钊、赵书梵 到课, 高健桓、蒋叔璋 线上

上周作业检查

上周作业链接: https://cppoj.kids123code.com/contest/3842

王向东老师周日三点半C++完全背包+多重背包

刷新

#	用户名	姓名	编程分	时间	A (12 / 21)	B (12 / 15)	C (12 / 15)	D (11 / 15)	E (8 / 11)	F (6 / 19)	G (4 / 6)
1	liuqi	刘祺	7	7:32:40	0:8:49 ♥	0:16:53 ♥	0:32:47 ♥	1:1:9 ♥	1:30:8	1:42:16	(-1) 2:0:38 ♥
2	zhaoshufan	赵书梵	7	7:54:42	0:10:4	0:43:30	0:51:36	1:18:51	1:17:22 ♥	1:24:30 ♥	2:8:49
3	yangyongcheng	杨咏丞	7	186:28:55	(-3) 0:17:0	0:41:32	0:54:58	1:33:35	1:53:33	(-4) 53:2:10	(-1) 125:26:7
4	liuliwei	柳力玮	7	211:32:41	0:11:32	(-1) 0:35:20	0:47:10	(-1) 1:16:18	(-2) 1:29:34	(-9) 101:20:8	101:32:39
5	tianxinyi	田心一	6	10:31:55	0:11:38	0:17:18	(-2) 0:33:50	(-1) 1:18:42	1:52:42	5:17:45	
6	gaojianhuan	高健桓	6	34:58:26	(-1) 0:14:9	0:29:2	0:45:40	1:22:41	1:59:4	29:47:50	
7	chujinyang	初锦阳	5	4:48:56	0:15:32	0:30:52	0:45:11	1:16:26	2:0:55		
8	yuanzhao	苑钊	5	5:40:15	(-1) 0:10:7	0:42:2	0:54:47	1:36:6	1:57:13		
9	jiangshuzhang	蒋叔璋	4	3:26:10	0:15:36	0:38:6	0:55:4	1:37:24			
10	jibohan	纪博涵	4	4:45:39	(-1) 0:14:32	(-1) 0:45:40	(-1) 1:11:26	1:34:1	(-1)		
[illegible]	liuchenxi	刘宸熙	4	5:8:56	(-1) 0:27:55	(-1) 0:38:1[illegible]	0:54:38	(-2) 1:48:6			
12	liruihan	李瑞涵	3	2:53:13	(-2) 0:13:14	0:43:26	1:16:33				

本周作业

https://cppoj.kids123code.com/contest/4033 (课上讲了 A ~ F 题, 课后作业是 F 题必做)

课堂表现

今天课上的 E 题是一个 bool 类型的 dp, 维护从 n 个数中选, 能得到哪些合法值

这道题课上很多同学都做的不好, 需要课后再复习复习。

课堂内容

[USACO08DEC] Hay For Sale S

裸 01 背包

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 5e4 + 5;
```

```
bool f[maxn];

int main()
{
  int V, n; cin >> V >> n;
  f[0] = true;
  while (n -- ) {
    int v; cin >> v;
    for (int i = V; i >= v; --i) f[i] |= f[i-v];
  }
  for (int i = V; i >= 0; --i) {
    if (f[i]) { cout << i << endl; break; }
  }
  return 0;
}
```

[USACO3.1] 总分 Score Inflation

裸完全背包

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 1e4 + 5;
int f[maxn];

int main()
{
  int V, n; cin >> V >> n;
  while (n -- ) {
    int c, v; cin >> c >> v;
    for (int i = v; i <= V; ++i) f[i] = max(f[i], f[i-v]+c);
  }
  cout << f[V] << endl;
  return 0;
}
```

榨取kkksc03

裸二维 01 背包

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 200 + 5;
int f[maxn][maxn];
```

```
int main()
{
  int n, M, T; cin >> n >> M >> T;
  while (n -- ) {
    int m, t; cin >> m >> t;
    for (int i = M; i >= m; --i) {
      for (int j = T; j >= t; --j) f[i][j] = max(f[i][j], f[i-m][j-t]+1);
    }
  }
  cout << f[M][T] << endl;
  return 0;
}
```

[USACO08NOV] Buying Hay S

f[i]: 买 >=i 的干草时, 最少花多少元

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 5e4 + 5;
int f[maxn];

int main()
{
  int n, V; cin >> n >> V;
  memset(f, 0x3f, sizeof(f));
  f[0] = 0;
  while (n -- ) {
    int v, c; cin >> v >> c;
    for (int i = 0; i <= V; ++i) f[i] = min(f[i], f[max(0,i-v)]+c);
  }

  cout << f[V] << endl;
  return 0;
}
```

kkksc03考前临时抱佛脚

对 4 组分别做 01 背包, 看每一组能凑出来的最接近 sum/2 的数是多少

```
#include <bits/stdc++.h>

using namespace std;

int len[5];
const int maxn = 1200 + 5;
bool f[maxn];
```

```
int calc(int n) {
  memset(f, false, sizeof(f));
  f[0] = true;
  int sum = 0;
  while (n -- ) {
    int x; cin >> x; sum += x;
    for (int i = maxn-1; i >= x; --i) f[i] |= f[i-x];
  }

  for (int i = (sum+1)/2; i <= sum; ++i) {
    if (f[i]) return i;
  }
  return 0;
}

int main()
{
  for (int i = 1; i <= 4; ++i) cin >> len[i];

  int res = 0;
  for (int i = 1; i <= 4; ++i) res += calc(len[i]);
  cout << res << endl;
  return 0;
}
```

积木城堡

思想跟上道题一致, 一共分了 n 组, 分别对每一组做 01 背包的 dp, 记录每一组能凑出来哪些数, 最后找一个都能凑出来的最大值即可

```
#include <bits/stdc++.h>

using namespace std;

const int N = 10000 + 5, M = 100 + 5;
bool f[M][N];

void solve(int id) {
  f[id][0] = true;

  while (true) {
    int x; cin >> x;
    if (x == -1) break;
    for (int i = N-1; i >= x; --i) f[id][i] |= f[id][i-x];
  }
}

int main()
{
  int n; cin >> n;
```

```
    for (int i = 1; i <= n; ++i) solve(i);

    for (int i = N-1; i >= 0; --i) {
      bool flag = true;
      for (int j = 1; j <= n; ++j) {
        if (!f[j][i]) flag = false;
      }
      if (flag) { cout << i << endl; break; }
    }
    return 0;
  }
```